

Divergence

If the vector field represents the velocity of a fluid or gas, then the **divergence** of the field is a measure of the “**compressibility**” of the fluid.

If the divergence is negative at a point, it means that the fluid is compressing: more fluid is going into the point than is going out.

If the divergence is positive, it means the fluid is expanding: more fluid is going out at that point than going in.

A divergence of zero means the same amount of fluid is going in as is going out. If the divergence is zero at all points, we say the field is **incompressible**.

$$\text{div } \vec{F} = \nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

The divergence of the vector is a scalar quantity.

Curl of a vector field

- **Curl** is a measure of the spinning action of the field.
- No spin corresponds to zero curl.
- The counter clockwise spin corresponds to positive curl and clockwise spin corresponds to negative curl.

$$\nabla \times \vec{V} = \xi = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix}$$

$$\nabla \times \vec{V} = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \hat{i} + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \hat{j} + \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \hat{k}$$

Curl of the gradient of a scalar function is zero.

$$\nabla \times \nabla \phi = 0$$